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Part 1: Planning The Mobile Application

Restaurant Menu Manager

1. Scenario Analysis

The restaurant currently relies on paper records to manage menu information. This makes it difficult for the chef to update dishes, organize menu items and quickly locate information. The proposed Restaurant Menu Manager application will provide a simple Android mobile solution for managing the restaurant's menu.

Requirements	Response
Who will use the application?	The application will mainly be used by Christoffel, the restaurant owner/chef, to manage the restaurant's menu from an Android mobile device.
What is the purpose of the application?	The purpose of the application is to provide a simple and efficient way for the chef to add, view, update, search, filter and manage menu items without relying on paper records. The purpose of the application is to provide a simple and efficient way for the chef to add, view, update, search, filter and manage menu items without relying on paper records.
What information must the application store?	The application must store the dish name, description, course and price for each menu item. The course can be Starter, Main Course or Dessert.
Four functions the application should perform.	1. Add new menu items. 2. View existing menu items. 3. Update and delete menu items. 4. Search and filter menu items.
Two usability requirements the application should have.	1. The application should have a simple, clear and easy-to-navigate interface so the chef can quickly manage dishes. 2. The application should provide clear labels, readable text and appropriate buttons, with validation to reduce input errors.

Main Application Requirements.

The application should support the following requirements:

- Add new menu items.
- View existing menu items.
- Edit existing menu items.
- Delete menu items.
- Search for menu items by dish name.
- Filter menu items by course.
- Display useful menu statistics.
- Provide clear navigation between screens.
- Validate information before saving a menu item.
- Provide a simple and user-friendly mobile interface.

2. User Interface Design

The proposed application will contain four main screens:

1. Home
2. Menu Items
3. Add Menu Item
4. Edit Menu Item

The interface will use a consistent design throughout the application. The same colours, fonts, spacing and button styles will be used on each screen to make the application easier to understand and use.

2.1 Screen 1: Home

The home will be the first screen displayed when application starts

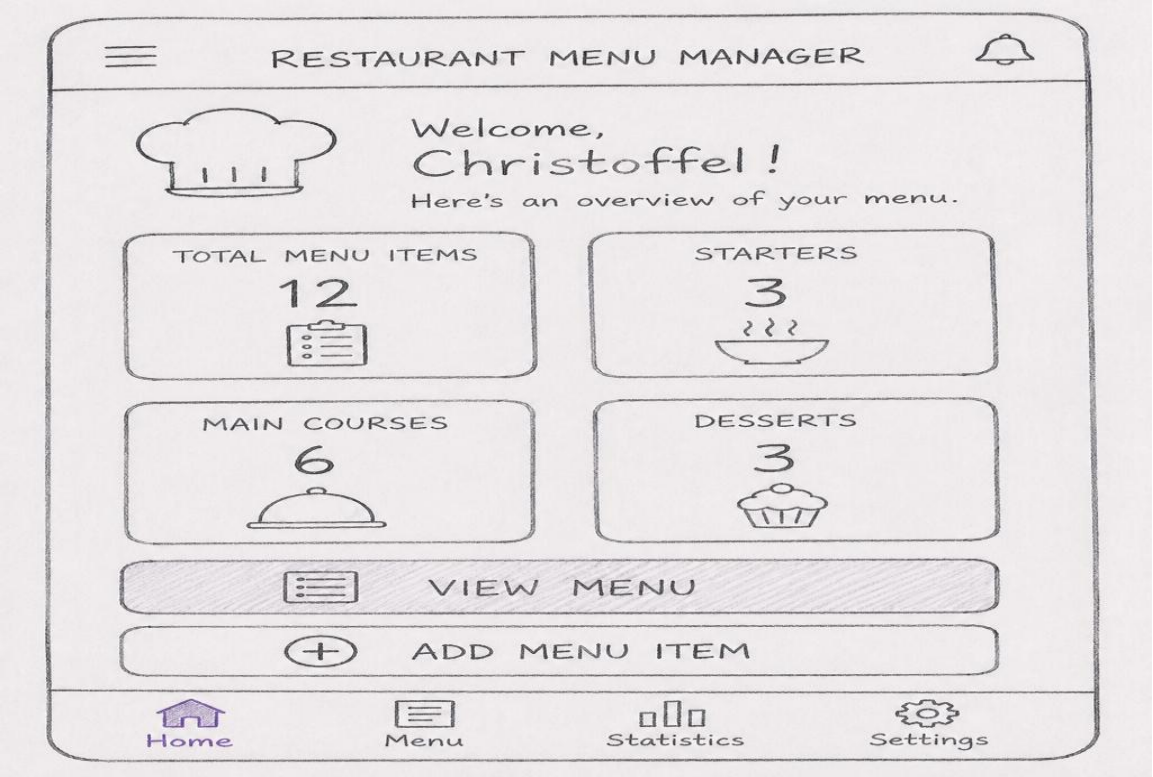
Purpose

The Dashboard provides an overview of the restaurant's menu and gives the chef quick access to the main functions of the application.

Interface Elements

- Application title: **Restaurant Menu Manager**
- Welcome message: **Welcome, Christoffel!**
- Total Menu Items statistic
- Starters statistic
- Main Courses statistic
- Desserts statistic
- View Menu button
- Add Menu Item button

Wireframe



The statistics shown above are examples for the wireframe and will be calculated from the actual menu data when the application is developed.

2.2 Screen 2: Menu Items

The Menu Items is the main menu management screen.

Purpose

This screen allows the chef to view, search and filter all menu items. The chef can also select an item to edit or delete it.

Interface Elements

- Screen title: Menu Items
- Search field
- Course filter
- Menu item list
- Edit button
- Add Menu Item button

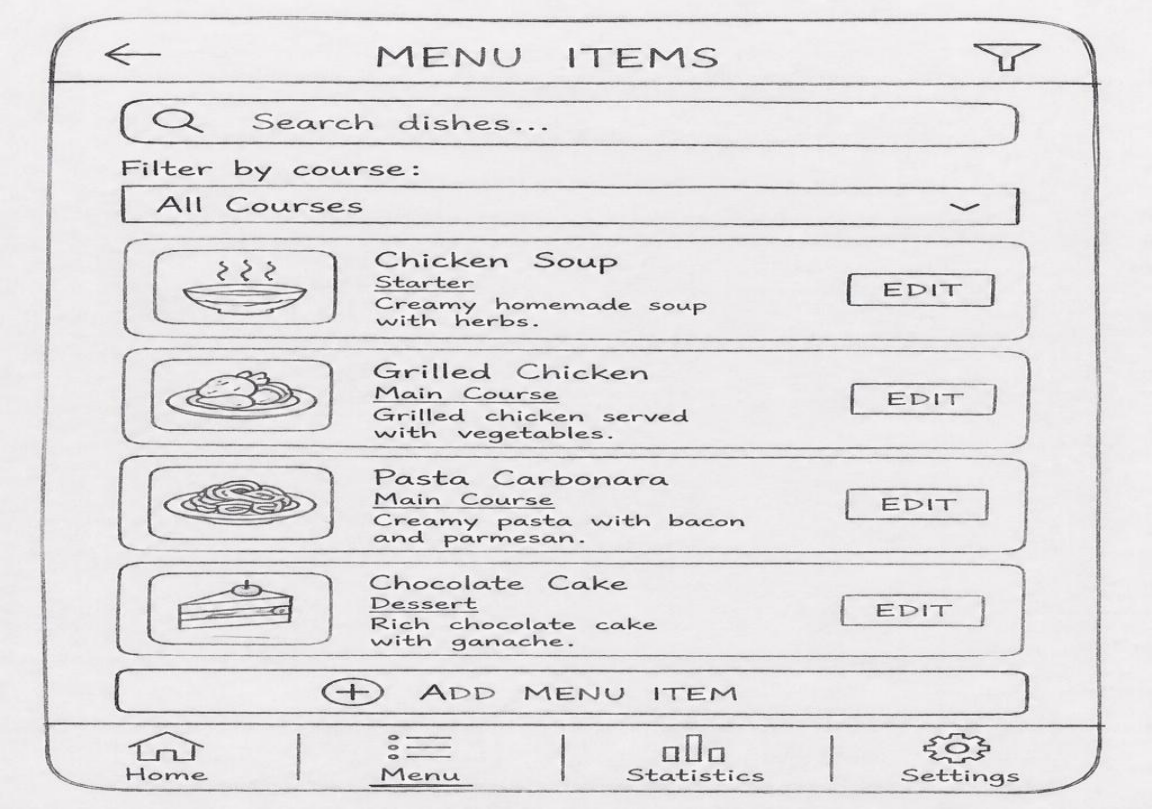
Course Filter Options

The course filter will contain:

- All
- Starter

- Main Course
- Dessert

Wireframe



The search field will allow the chef to enter part or all of a dish name. The course filter will allow the chef to display only dishes belonging to a selected course.

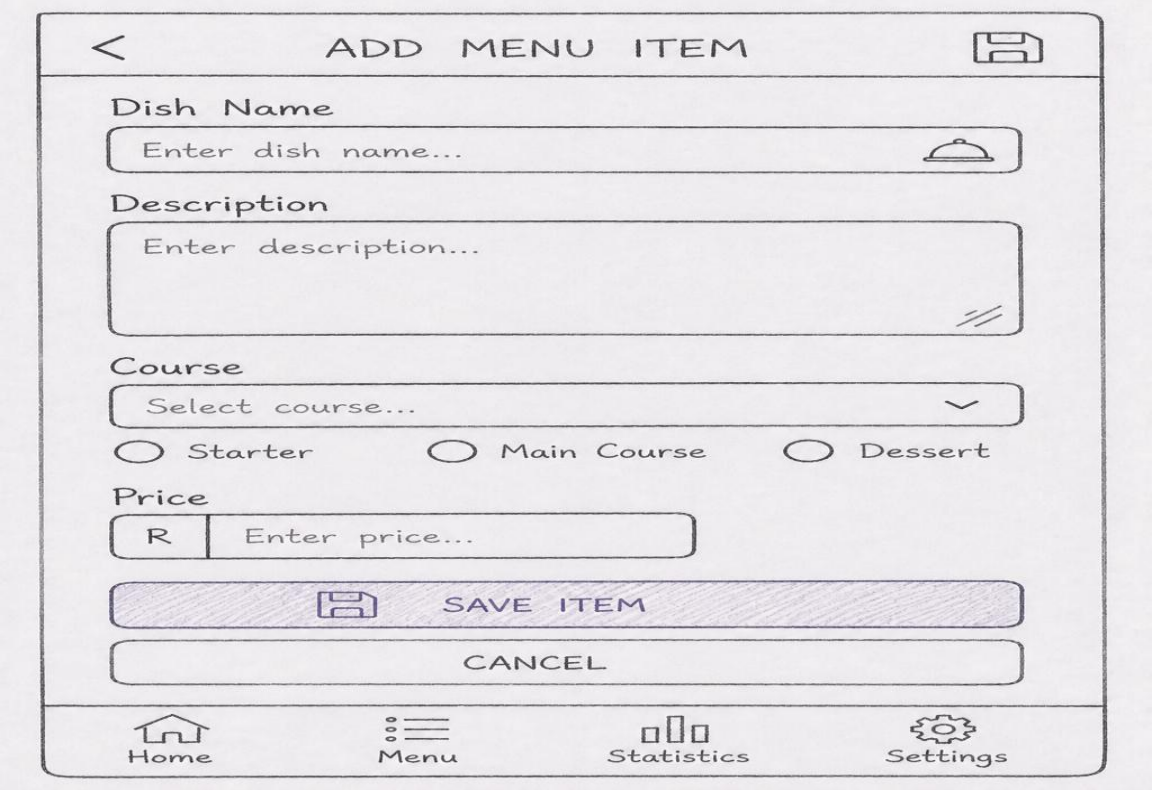
2.3 Screen 3: Add Menu Item

The Add Menu Item screen allows the chef to create a new menu item.

Interface Elements

- Screen title: **Add Menu Item**
- Dish Name text field
- Description text field
- Course picker
- Price text field
- Save Item button
- Cancel/Back button

Wireframe



The application will validate the required fields before saving the menu item. For example, the dish name and course should not be empty, and the price should contain a valid numerical value.

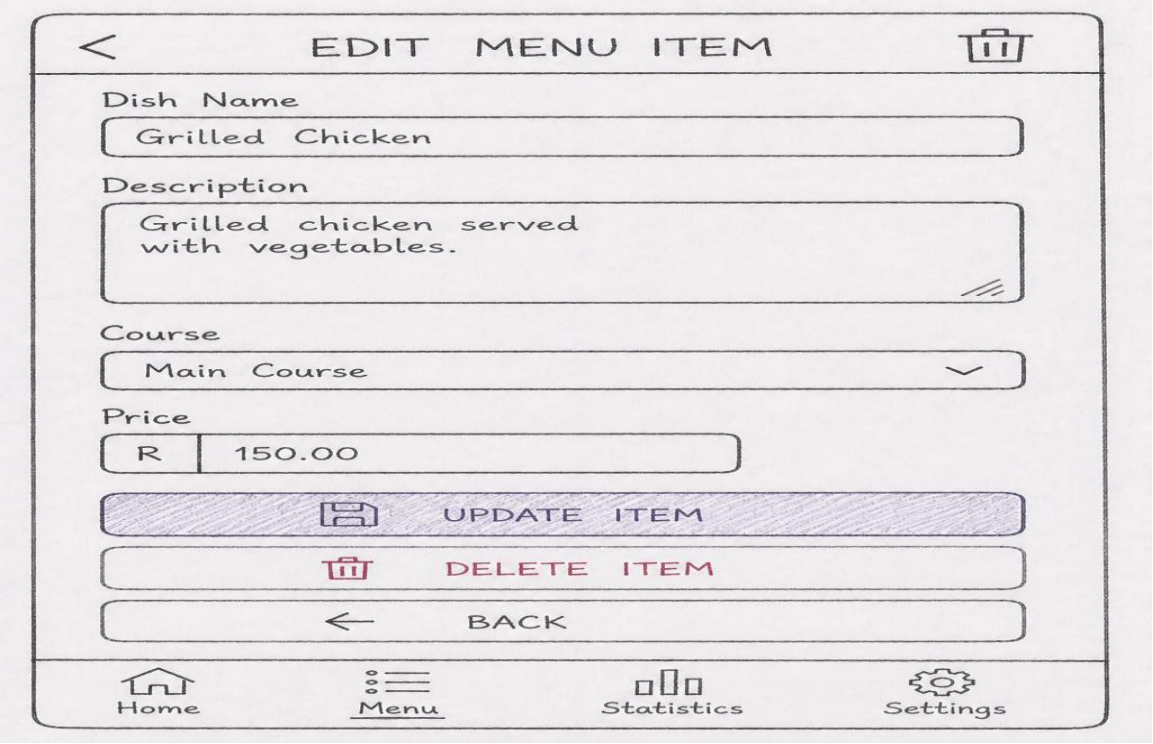
2.4 Screen 4: Edit Menu Item

The Edit Menu Item screen will have a similar layout to the Add Menu Item screen. The difference is that the existing menu item's information will already be displayed in the fields.

Interface Elements

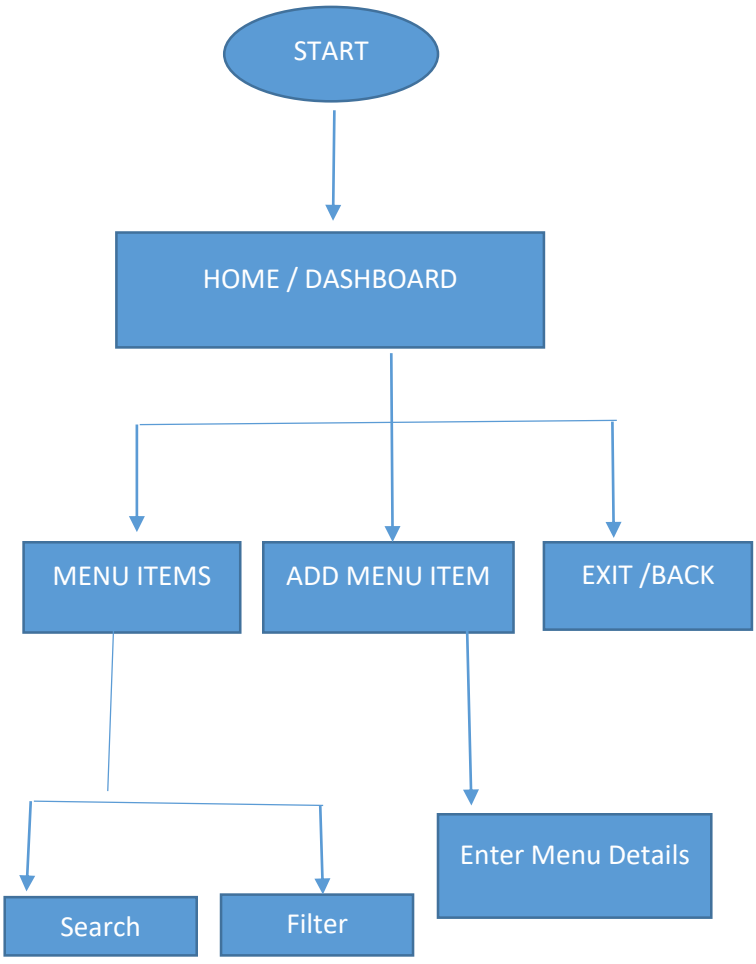
- Screen title: **Edit Menu Item**
- Populated Dish Name field
- Populated Description field
- Course picker
- Populated Price field
- Update Item button
- Delete Item button
- Back button

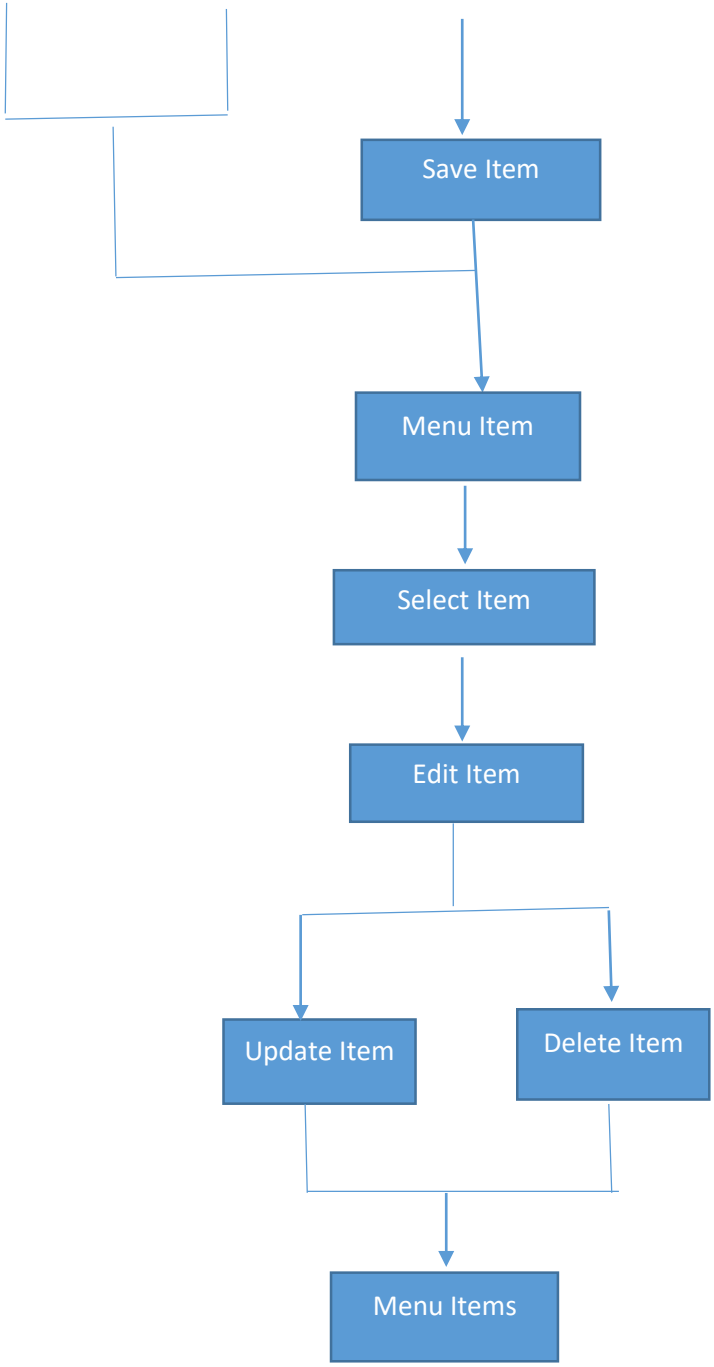
Wireframe



When the chef selects Update Item, the changes will be saved. When Delete Item is selected, the application should request confirmation before permanently removing the item.

3. Application Flow





The application flow demonstrates how the chef will move between screens and complete important actions.

Application Flow Explanation

The application starts on the **Home/Dashboard** screen. From the Dashboard, the chef can either view the menu items or add a new menu item.

When the chef selects **Menu Items**, all stored menu items are displayed. The chef can search for a specific dish using the search field or filter the menu according to course.

Selecting an existing menu item allows the chef to access the **Edit Menu Item** screen. From this screen, the chef can update the item's information or delete it.

When adding a new item, the chef enters the dish name, description, course and price. After selecting **Save Item**, the new menu item is added and the chef is returned to the Menu Items screen.

This flow provides a simple path between the main screens while supporting the important menu management operations.

4. React Native Component Planning

The following React Native components have been selected based on the requirements of the application.

Application Requirement	React Native Component	Purpose
Display a heading	Text	Displays screen titles, headings and other text.
Enter the dish name	TextInput	Allows the chef to enter the name of a dish.
Enter the description	TextInput	Allows the chef to enter a description of the dish.
Select a course	Picker	Allows the chef to select Starter, Main Course or Dessert.
Enter the price	TextInput	Allows the chef to enter the price of the menu item.
Save a menu item	Button / Pressable	Allows the chef to save the entered menu item.
Display menu items	FlatList	Efficiently displays multiple menu items in a scrollable list.

Additional Components

Additional Requirement	React Native Component	Purpose
Search menu items	TextInput	Allows the chef to search for a specific dish.
Filter by course	Picker	Allows the chef to display dishes from a selected course.
Navigate between screens	Pressable / Button	Allows the chef to interact with navigation controls.
Organise screen layout	View	Groups and arranges interface elements.
Display confirmation messages	Alert	Provides confirmation and warning messages, such as before deleting an item.
Display scrollable content	ScrollView	Allows content to be viewed when it exceeds the available screen space.

Component Selection Rationale

Text is appropriate for headings, labels and menu information**. TextInput** is suitable for user-entered information such as dish names, descriptions and prices. A **Picker** provides a controlled selection for the three available course categories. **Button** or **Pressable** components provide clear interactive actions.

FlatList is particularly appropriate for the Menu Items screen because it is designed to display collections of data efficiently. **View** will be used to organise the layout of each screen, while **Alert** can be used to confirm potentially destructive actions such as deleting menu items.

For the React Native implementation, the course picker can be implemented using the React Native Picker component/library appropriate to the project's React Native setup.

5. Design Reflection

The proposed Restaurant Menu Manager application is designed to make menu management easier, faster and more organised for Christoffel. The current paper-based system makes it difficult to update dishes and quickly find menu information. My proposed application addresses these problems by providing a simple mobile interface where menu items can be added, viewed, updated, deleted, searched and filtered.

The application will use a Dashboard as the starting screen. The Dashboard will display useful menu statistics, including the total number of menu items and the number of Starters, Main Courses and Desserts. A separate Menu Items screen will allow the chef to search for dishes and filter the list according to course. This will make it easier to find specific dishes without having to search through paper records.

The Add and Edit screens will contain clearly labelled input fields for the dish name, description and price, together with a course picker. This layout keeps the process of adding or changing a menu item simple and straightforward. The interface will use consistent buttons, readable text and clear navigation to improve usability. Validation will also help reduce incorrect or incomplete information.

The selected React Native components contribute to a user-friendly application. **Text** and **TextInput** components will display information and allow the chef to enter data. A Picker will provide an easy way to select the course, while **Button** or **Pressable** will allow actions such as saving and updating items. **FlatList** will display menu items in an organised and efficient list.

Overall, the design focuses on simplicity, clarity and efficiency while meeting the chef's menu management requirements.

6. References

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